



BATTERIES FROM NUCLEAR WASTE

I thought the 'Batteries From Nuclear Waste That Last Thousands Of Years' article published in November 2020 issue was a great piece and exciting technology. However, I'm shocked to hear there's a shortage of potential fuel in the USA. We hear the opposite, and the controversy of where to store spent nuclear waste.

James Copeland

❑ I liked this article. Good and simple information. Became a little lengthy but good!

Dipro Sarkar

❑ Disposal of nuclear waste in such batteries does make sense. Probably it will open ways of making long lasting batteries in future.

Jai Singh Baghel

❑ This is the future. May be in hundred years we will power all electronic devices with lifetime batteries. Bravo!

Ivana TGTG

❑ It is a nice article. Thank you!

Hasan

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ARTICLES IN MARCH ISSUE

I would like to comment on some of the articles published in March issue.

- Presently, due to increase in excise duty, the prices of electronic goods are likely to increase but in the long run we will have better indigenous electronic goods.

- The design of crystal radio is quite simple but its antenna realisation is a tedious task.

- The autonomous vehicles are interesting but their realisation and

availability for regular use without a driver will take some more time.

O. Subrahmanyam

Retired Scientist, DRDO, Visakhapatnam

EFY. Thanks to everyone for the feedback!

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FM TRANSMITTER

The specification of 'Make A Crystal-Locked FM Transmitter' DIY article published in April 2020 issue is dreadful. The spurious products are only -20dBc. Proper transmitters have spurs in the range of -70dBc to -90dBc.

Albert H.

The author Joy Mukherji replies:

This circuit was aimed at an average hobbyist. To keep things simple, there is only a single variable capacitor for tuning. It will work the first time, if built properly, and has a decent range.

The distortion is low enough for voice. It illustrates the simple technique of modulating a crystal for narrow-band frequency modulation. This technique has been used in VHF amateur radio transceivers as well as for 2-metre band ham radio.

My main design goals were rock solid frequency stability, a usable output power with common 2N3904 transistors, and simplicity.

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AUTOMATIC HAND-WASH TIMER

In the 'Automatic Hand-Wash Timer' DIY project published in July 2020 issue, how does it turn on for twenty seconds? Is capacitor C2 (100µF) of fixed value?

Arjun S.

EFY. Resistor R3 and capacitor C2 are timing components. The given values of these components decide the 'on' duration of around twenty seconds. Capacitor C2 used in the circuit is not variable. However,

you can replace it with a variable capacitor if you want to change the output 'on' time.

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AUTOMATIC HAND-WASH TIMER

I have assembled the 'Automatic Hand-Wash Timer' DIY project published in July 2020 issue, as shown in the circuit diagram, but it is not working. When I checked it using multimeter, the current is flowing in the circuit. Please help me.

Muhammed Sinan

EFY. Problem could be loose connections in the circuit. For testing, first remove the IR sensor and short pin 2 of NE555 to ground using a wire and then disconnect the wire. You should see the LED glow for twenty seconds. If the LED does not glow, check your connections properly. If you have assembled the circuit on PCB, check dry soldering and, if necessary, reheat the solder joint(s) using a good soldering iron.

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HI-FI AUDIO AMPLIFIER

This is regarding '14-Watt Hi-Fi Audio Amplifier Using TDA2030A' DIY project published in September 2020 issue. I want to build this project on the PCB. Kindly inform about the availability of its PCB, components, and other accessories used in the project.

Rahit Kumar Roy

EFY. The PCB is not available with us right now. You can easily assemble the circuit either on a Veroboard (general-purpose PCB) or on a breadboard. The rest of the components, including Veroboard, can be purchased from kitsnspares.com (or you may contact them at info@kitsnspares.com).

Note. Most articles published in EFY magazine can also be read at our website <https://www.electronicshobby.com> after some time, including DIY articles whose components can be purchased from our kitsnspares.com website (or by writing to info@kitsnspares.com).